1. Show that the series $\displaystyle\sum_{k=1}^{\infty} \frac{z^k}{k}$ converges on $\{|z| \leq 1\}$ except at $z = 1$.

2. Suppose that f is holomorphic in a region and that, at every point, either $f = 0$ or $f' = 0$. Show that f is a constant.

3. Prove that a nonconstant holomorphic function cannot map an open region into a straight line or into a circular arc.

4. Let $U = \{z \in \mathbb{C} \mid |z| < 1\}$ be the open unit disc. For every $a \in U$, define $\varphi_a(z) = \dfrac{a - z}{1 - \overline{a}z}$ on U. Show that $\varphi_a \in \mathrm{Aut}(U)$, the automorphism group of U, i.e., an automorphism of U is a holomorphic map from U into itself which is one-to-one and onto. Show also that φ_a maps ∂U one-to-one and onto ∂U.

5. Let $g(z)$ be an entire function with $\operatorname{Im} g(z) \leq 0$. Show that g is a constant function.

6. Suppose that f is an entire function satisfying $|f(z)| \leq \dfrac{1}{|\operatorname{Im} z|}$ for all z. Prove that $f \equiv 0$.

7. Suppose $P(z) = a_0 + a_1 z + \cdots + a_n z^n$ is bounded by 1 for $|z| \leq 1$. Show that $|P(z)| \leq |z|^n$ for all $|z| \geq 1$.

8. Let g be an entire function such that $|g(z)| \leq A + B|z|^k$, where $k > 0$, $A > 0$, $B > 0$. Show that g is a polynomial with degree less than or equal to k.

9. Find the sum of the distances from the point 1 to the other nth roots of 1. Divide the result by n and let $n \to \infty$ to conclude that the average distance from 1 to a point on $|z| = 1$ is $4/\pi$.

10. Prove Lagrange's identity:

$$\left|\sum_{k=1}^{n} z_k w_k\right|^2 = \left(\sum_{k=1}^{n} |z_k|^2\right)\left(\sum_{k=1}^{n} |w_k|^2\right) - \sum_{k\leq j} |z_k\overline{w_j} - z_j\overline{w_k}|^2 .$$

11. Apply the Residue Theorem to evaluate $\displaystyle\int_0^\infty \frac{dx}{1+x^m}$, where $m \geq 2$ is a positive integer. (Hint: Use the boundary of a sector with angle $\dfrac{2\pi}{m}$ as the contour.)

12. Apply the Residue Theorem to evaluate $\displaystyle\int_0^\infty \frac{x^2}{1+x^{10}}\,dx$. (See the hint of exercise 11.)

13. Apply the Residue Theorem to evaluate $\displaystyle\int_0^\infty \frac{\sin^2 x}{x^2}\,dx$. (Hint: consider $\displaystyle\frac{e^{2iz}-1-2iz}{z^2}$.)

14. Suppose f is an entire function and $|f(z)| = 1$ on $|z| = 1$. Prove that $f(z) = e^{i\theta_0} z^m$, for some $\theta_0 \in \mathbb{R}$ and $m \in \mathbb{N}$.

15. Show that if f is meromorphic in the extended plane, then f is a rational function.

16. Suppose that f is holomorphic inside and on a simple closed curve C and has no zeros on C. Show that, if m is a positive integer, then

$$\frac{1}{2\pi i}\int_C z^m \frac{f'(z)}{f(z)}\,dz = \sum_k z_k^m,$$

where the sum is taken over all the zeros $\{z_k\}$ of f inside C.

17. Show that there is no non-constant holomorphic function in the open unit disc which is continuous up to the unit circle and is real on the unit circle.

18. Suppose f is an entire function such that $f(0) = 0$ and $\operatorname{Re} f(z) \to 0$ as $|z| \to \infty$. Prove that $f \equiv 0$.

19. Let Ω be a bounded domain containing 0 in $\mathbb{C}$, and let f be a holomorphic function on Ω that maps Ω into itself. Suppose also that $f(0) = 0$ and $f'(0) = 1$. Show that $f(z) \equiv z$.

20. Is it possible to approximate $1/z$ uniformly by polynomials on the annulus $A = \{z \mid \frac{1}{2} < |z| < 1\}$? Show your reason.

21. Evaluate $\displaystyle\int_0^\infty \frac{x^\lambda}{a^2+x^2}\,dx$, $-1<\lambda<1$, $a>0$.

22. Evaluate $\displaystyle\int_0^\infty \frac{\ln x}{x^\lambda(1+x)}\,dx$, $0 < \lambda < 1$.

23. Let f be a holomorphic function defined on the open unit disc such that $\left|f\left(\frac{1}{n}\right)\right| \leq \frac{1}{3^n}$ for $n \geq 2$. Prove that f is identically zero.

24. Let $p(z) = a_0 + a_1 z + a_2 z^2 + \cdots + a_n z^n$ be a polynomial with all a_j real and $0 \leq a_0 \leq a_1 \leq \cdots \leq a_n$. Show that all of the zeros of $p(z)$ lie inside the closed unit disc.

25. Let $p(z) = 1 + 2z - 18z^4$. Show that all the zeros of p lie within the open disc $D = (0; \frac{2}{3})$.

26. Show that the only univalent entire functions are the affine functions $f(z) = az + b$, $a, b \in \mathbb{C}$, $a \neq 0$.

27. Suppose g is holomorphic in the punctured plane $z \neq 0$ and satisfies $|g(z)| \leq \sqrt{|z|} + \dfrac{1}{\sqrt{|z|}}$ for all $z \neq 0$. Prove g is a constant.

28. Let $\{m_1, m_2, \ldots, m_k\}$ be a set of positive integers and

$$R(z) = \frac{1}{(z^{m_1} - 1)(z^{m_2} - 1) \cdots (z^{m_k} - 1)}.$$

Find the coefficient c_{-k} in the Laurent expansion for $R(z)$ about the point $z = 1$.

29. Let $g \in \mathcal{O}(\Omega)$, where $\Omega = U \setminus \{0\}$. Suppose that $\iint_\Omega |g(z)|^2 \, dx \, dy < \infty$. Show that 0 is a removable singularity of g.

Here U denotes the open unit disc in $\mathbb{C}$.

30. Show that the converse of Darboux-Picard's theorem is false: Find a simple closed curve C and a function f which is holomorphic on and inside C such that f is univalent inside C but not on C.

31. Suppose that g is entire and $g(z)$ is real if and only if z is real. Show that g can have at most one zero.

32. Suppose $f \in \mathcal{O}(U)$. Prove that there is a sequence $\{z_n\}$ in U such that $|z_n| \to 1$ and $\{f(z_n)\}$ is bounded.

Here U denotes the open unit disc in $\mathbb{C}$.

33. Let $f \in \mathcal{O}(\overline{U})$. If $|f(z)| \leq 1$ for $|z| = 1$ and $f(0) = \frac{1}{2}$. Show that $|f(z)| \leq \dfrac{3|z|+1}{2}$ for all $z \in U$.

Here U denotes the open unit disc in $\mathbb{C}$.

34. Let $f \in \mathcal{O}(\overline{U})$, and $|f(z)| < 1$ for $|z| = 1$. How many fixed points must f have in the disc?

Here U denotes the open unit disc in $\mathbb{C}$.

35. Let K be a compact set in $\mathbb{C}$, and let $g(z) = az + 1$, $a \in \mathbb{C}$. Suppose that the origin lies in the convex hull of K (the smallest closed convex set containing K). Show that $\sup_{z \in K} |g(z)| \geq 1$. Note that 0 may not lie in K.

36. Find holomorphic function $f(z)$ whose real part is $x^3 - 3x^2y - 3xy^2 + y^3$.

37. Write out Poisson's integral formula on a disc $D(a;R) = \{z \mid |z-a| < R\}$, if u is continuous on $\overline{D(a;R)}$ and harmonic in $D(a;R)$.

38. Let Ω be a region, and let $f_n \in \mathcal{O}(\Omega)$, $n \in \mathbb{N}$. Suppose u_n is the real part of f_n and $\{u_n\}$ converges uniformly on compact subsets of Ω and $\{f_n(z)\}$ converges for at least one point in Ω. Prove that $\{f_n\}$ converges uniformly on compact subsets of Ω.

39. If u is real-valued and harmonic on a connected open set, and if u^2 is also harmonic, prove that u is a constant function.

40. Show that the Laplace operator $\dfrac{\partial^2}{\partial x^2} + \dfrac{\partial^2}{\partial y^2}$, when written in polar coordinates, takes the form

$$\Delta = \frac{1}{r}\frac{\partial}{\partial r}\left(r\frac{\partial}{\partial r}\right) + \frac{1}{r^2}\frac{\partial^2}{\partial \theta^2}.$$

41. Let u be a bounded real-valued harmonic function in U. Is the conjugate harmonic of u also bounded in U? Prove it or give a counterexample.

Here U denotes the open unit disc in $\mathbb{C}$.

42. Let f be an entire function, and let R be a rectangle. If $f(R)$ is also a rectangle, prove that f is linear.

43. If g is an entire function with the property that $|g(z)| \to \infty$ as $|z| \to \infty$, prove that $g(\mathbb{C}) = \mathbb{C}$.

44. Suppose $f \in \mathcal{O}(U)$, and there is a constant $M > 0$ such that $|f^{(k)}(0)| \leq M^k$ for all k. Show that f can be extended holomorphically to an entire function.

Here U denotes the open unit disc in $\mathbb{C}$.

45. Let f be holomorphic and bounded on $|z+i| > \frac{1}{2}$ and real on $(-1,1)$. Show that f is a constant function.

46. Characterize $\mathrm{Aut}(U \setminus \{0\})$.

Here U denotes the open unit disc in $\mathbb{C}$.

47. Find the image of $\Omega = \{z \in \mathbb{C} \mid \operatorname{Re} z < 0,\ |\operatorname{Im} z| < \pi\}$ under the exponential function.

48. Let $\{f_\alpha\}_{\alpha\in\Lambda}$ be a normal family of holomorphic functions on a domain D. Prove that $\{f'_\alpha\}_{\alpha\in\Lambda}$ is a normal family.

49. Let $\mathcal{F}$ be the class of all $f \in \mathcal{O}(U)$ such that $\operatorname{Re} f > 0$ and $f(0) = 1$. Is $\mathcal{F}$ a normal family?

Here U denotes the open unit disc in $\mathbb{C}$.

50. Let $\mathcal{F}$ be the class of all $f \in \mathcal{O}(U)$ for which

$$\iint_U |f(z)|^2 \, dx \, dy \leq 1.$$

Is $\mathcal{F}$ a normal family?

Here U denotes the open unit disc in $\mathbb{C}$.

51. Given an entire function which is real on the real axis and imaginary on the imaginary axis, prove that it is an odd function, i.e., $f(z) = -f(-z)$.

52. Characterize $\mathrm{Aut}(D)$, where $D = U \setminus \{(\frac{1}{3}, 0), (-\frac{1}{4}, 0)\}$.

Here U denotes the open unit disc in $\mathbb{C}$.

53. Let $a > 0$, and define $C = \left\{ z \in \mathbb{C} \,\middle|\, \dfrac{|z-a|}{|z+a|} = \rho \right\}$, for some $\rho > 0$. Show that (i) C is a circle, and (ii) a and $-a$ are symmetric with respect to C.

54. Show that a Möbius transformation T satisfies $T(0) = \infty$ and $T(\infty) = 0$ if and only if $T(z) = \dfrac{a}{z}$ for some nonzero a in $\mathbb{C}$.

55. Let $a, b \in \mathbb{R}$, $a < b$, and let $S = \{x + iy \mid a < x < b \text{ and } y > 0\}$. Find a biholomorphic mapping f from S onto U.

Here U denotes the open unit disc in $\mathbb{C}$.

56. Find a holomorphic function that maps the complement of the line segment $[-1, 1]$ into the open unit disc U.

57. Show that $g(z) = 2z + \dfrac{1}{z}$ maps the exterior of the unit circle conformally onto the exterior of the ellipse: $\dfrac{x^2}{9} + y^2 = 1$.

58. Let Ω be an open connected set bounded by two circles which are tangent to each other at one common boundary point a. Map Ω conformally onto the open unit disc U.

59. Let Ω be a domain bounded by two circles $C_1 = \{z \mid |z| = 1\}$ and $C_2 = \{z \mid |z - \frac{1}{2}| = \frac{1}{4}\}$. Map Ω conformally onto an annulus.

60. Let $|a| < R$. Show that

$$f(z) = \frac{R(z-a)}{R^2 - \overline{a}z}$$

is holomorphic on $|z| \leq R$, and maps the circle $|z| = R$ into the unit circle.